Ch 5 > Economic Analysis

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- MARR ($i\%$) → Minimum Attractive Rate of Return
  - → Minimum Return to invest
  - → factors
    - → Investment Amount
    - → Nature of Investment
    - → Type of Organization

- Economic Analysis Method

A. Equivalent Worth
- → PW
- → AW
- → FW

↳ EW > 0 : Best

A1. PW (Present Worth)

$$PW = -I + An(P/A, i\%, n) + S(P/F, i\%, n)$$

A2. FW (Future Worth)

$$FW = -I(F/P, i\%, n) + An(F/A, i\%, n) + S$$

A3. AW (Annual Worth)

$$AW = \text{Revenue} - \text{Expense} - CR$$

$$CR = -I(A/P, i\%, n) + S(A/F, i\%, n)$$

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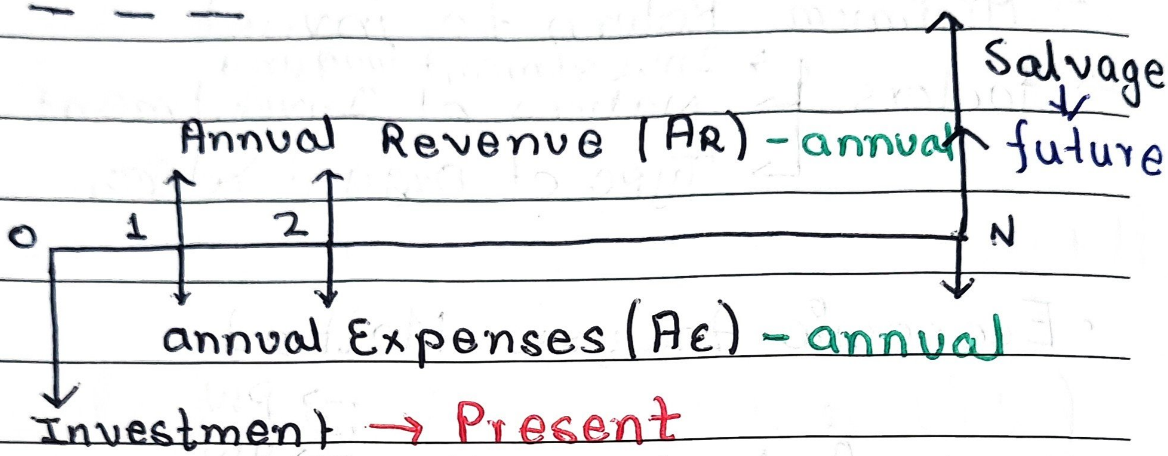

$$F = P(1+i)^N = A\left(\frac{(1+i)^N - 1}{i}\right)$$

— — —

* if n = diff, use LCM(n, m)

B. Payback Period

- time जसमा I फिर्ता हुन्छ

* $PP_{calculated} < P_{defined}$: Accept

B1. Simple Payback Period

$$PP = \frac{\text{investment}}{\text{savings}}$$

B2. Discounted Payback Period

$$PP = \frac{TVM\ I}{TVM\ S}$$ (TVM = Time value of money)

- - -

EOY	Net Cash flow		Cummulative CF
i	F_i	$\frac{F_i}{(1+i)^n}$	$F_i + F_{i-1} + \ldots + F_o$

$$PP = i\,|-ve\ CF| + \frac{CF_i}{F_{i+1}}$$

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C. Rate of Return (RR) ↳ IRR
↳ ERR

* RR > MARR : accept

C1. Internal Rate of Return (IRR)

PW inflow (i%) = PW outflow (i%)

Calculate i = IRR & decide

* Cashflow w/ EOY [देख] CFlow = F

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C2. External Rate of Return (ERR)

Given $\epsilon\%$ (Reinvestment rate)

1. Calculate : $PW_{outflow}$ ($\epsilon\%$)
2. Calculate : FW_{inflow} ($\epsilon\%$)
3. Solve i = ERR + decide

$$PW_{outflow}(1+i)^N = FW_{inflow}$$

* FW निकालदा n रवाल्न शरक्ने

D. Public Sector Economic Analysis

B/C Ratio | S/I Ratio

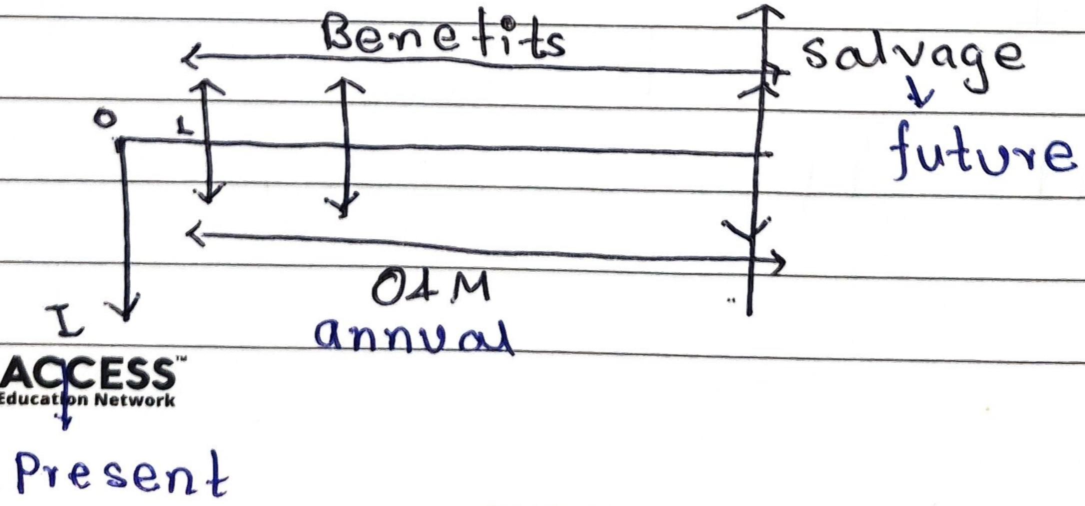
Benefit to cost | Saving to investment

Can be done by both : AW & FW

* $B/C > 1$: Accept

D1. Conventional $B/C = \dfrac{\text{Benefits}}{I + O\&M - S}$

D2. Modified $B/C = \dfrac{\text{Benefit} - O\&M}{I - \text{Salvage}}$



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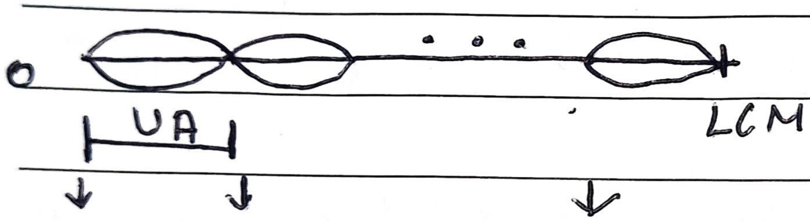
• Compairing mutually Exclusive project w/ unequal useful Life

1. Repeatability Assumption

- Repeat the project
  LCM(UA, UB)



PW

2. Coterminated Assumption

Case I) Study Period > Useful life

$$
\begin{array}{cccc}
 & UL & & SP \\
0 \vdash\!\!\!-\!\!-\!\!-\!\!-\!\!\dashv & & - - - \dashv & \\
 & FW_{UL} & & FW_{SP}
\end{array}
$$

$$FW_{SP} = FW_{UL}(1+i)^{SP-UL}$$

Case II) Study Period < Useful life

- Imputed Market Value (MV) calculation

$$MV = PW_{CR} + PW_{MV} \qquad (CR = -I + S)$$

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- Project Alternatives

A. Mutually Exclusive → Only 1 is selected

B. Independent → Choice of Project-A doesn't affect B

C. Dependent → choice of Project-A affect B

D. Contigent → Suru मा stg गारेसी बल्ल अर्को

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